SWE 4501 **Design Patterns** **Credit: 2.00**

Design patterns: design for reuse; capture and communication of knowledge and experience; pattern languages; kinds of patterns; choosing and using patterns; History of patterns: model-view-controller in Smalltalk; Alexander's patterns in architecture; Some common patterns: model-view-controller, observer, adapter, Façade, Layer, abstract factory, composite, command, iterator, visitor, proxy, strategy; Anti-patterns: bad situations and how to get out of them, development, architectural and managerial anti-patterns, recovery, refactoring and realignment; A case study: iterative development of an extended practical example; a case study in the application and use of patterns.

Recommended Texts:

1. E. Gamma et al., *Design Patterns: Elements of Reusable Object-Oriented Software*, 1st ed. Addison-Wesley Professional, 1994

SWE 4503 **Software Security** **Credit: 3.00**

Introduction to Software Security, Major security flaws, Types of threats, OS Security: Memory, CPU and I/O, Program security: String handling, Dynamic Memory, Input validation, and others, OWASP Listed Vulnerabilities, Concurrency and race condition, Best Practices: Secure programming guidelines, Security Standards, ways to avoid security holes in new software, methodologies and tools for identifying and eliminating security vulnerabilities, Scripting.

Recommended Texts:

1. C. P. Pfleeger et al., *Security in Computing*, 6th ed. Addison-Wesley Professional, 2023
2. W. Stallings and L. Brown, *Computer Security: Principles and Practice*, 4th ed. Pearson, 2017
3. B. Chess and J. West, *Secure Programming with Static Analysis*. Addison-Wesley Professional, 2007
4. D. A. Wheeler, *Secure Programming HOWTO*, 3.72th ed. Citeseer, 2015

Math 4544 **Numerical Methods Lab** **Credit: 0.75**

Sessional works based on CSE 4543.

CSE 4502 **Operating Systems Lab** **Credit: 1.00**

Sessional works based on CSE 4501.

SWE 4502 **Design Patterns Lab** **Credit: 1.00**

Sessional works based on SWE 4501.

SWE 4504 **Software Security Lab** **Credit: 0.75**

Sessional works based on SWE 4503.

SWE 4506 **Design Project I** **Credit: 1.50**

This will be a group project with three students in each group. It will test the ability to work as a member of a group. Each student of the group will have specific responsibilities. The duration of the project will be one year. In this course, students will focus on the design portion based on Software Requirements Specification (SRS) to implement a particular project.

Elective 5-I (Selective)

SWE 4531 **Network Programming** **Credit: 3.00**

Basic Networking Software (Protocol stacks, TCP/IP, HTTP, etc) Internet architecture and history, Elementary socket programming in C, Low level networking, Ethernet, ARP, The network layer, IP, DHCP, NAT, The network layer, routing, IPv6, Transport layer protocols, TCP, UDP, The socket interface (writing clients and servers) Advanced socket programming, nonblocking sockets, Server design (forking, threads, preforking), daemons, Network Programming in Java, DNS, email, HTTP, cgi, cookies, P2P Web services (XML, JSP, SOAP, etc) XML, DTDs, Schemas, XML Parsing, XSLT, Client side scripting, Javascript, AJAX, Web server technologies, Tomcat, servlets, Web server technologies, JSP, Web server, technologies, RPCs, Java RMI, XMLRPC, CORBA, Server scripting languages, PHP, Ruby Web services, SOAP, WSDL, UDDI, The Semantic Web, RDF, OWL Network security Cryptography, authentication, digital signatures, Network security, Kerberos, IPSec, SSL, Implementation of security, Anonymity on the Web, tor, Multimedia and VoIP, RTP.

Recommended Texts:

1. W. R. Stevens, *UNIX Network Programming*. Pearson, 1990